Practical Comparison: OLS, Ridge, and Lasso Regression

Overview

This handout summarizes when to use Ordinary Least Squares (OLS), Ridge Regression, and Lasso Regression, along with real-world advantages and limitations.

When to Use Each Method

Method	Best Used When	Pros	Cons
OLS (Linear Regression)	• Small number of predictors • Low multicollinearity • Focus on explanation & inference	• Simple and interpretable • Unbiased coefficient estimates	• Can overfit • Unstable with correlated predictors
Ridge Regression (L2)	• Many predictors • Predictors are correlated • Most variables influence the response	• Reduces overfitting • Handles multicollinearity • Improves prediction stability	• Does not remove variables (no sparsity) • Less interpretable than OLS
Lasso Regression (L1)	• Many predictors, only some important • Want automatic feature selection • High-dimensional problems ($p > n$)	• Produces simpler, sparse models • Selects important features • Improves interpretability	• Can drop useful variables • Biased estimates • Unstable when predictors highly correlated

Quick Guidelines

- Use **OLS** when the model is small and interpretation matters.
- Use **Ridge** when predictors are highly correlated or all variables should stay.
- Use **Lasso** when you expect only a subset of predictors to matter.
- When predictors are highly correlated and feature selection is desired, consider **Elastic Net**.
- Always tune λ using cross-validation.

Summary

- **OLS:** Best for inference, simple, but can overfit.
- **Ridge:** Shrinks coefficients, keeps all predictors, reduces variance.
- **Lasso:** Shrinks coefficients, removes irrelevant predictors, improves interpretability.